B.5 PUFFIN

This dataset can be found in the Huggingface hub under the name LDJnr/Puffin. Puffin contains 3000 conversations with GPT-4, each being a sequence of interactions that start with the human's query. We constructed two samples from this dataset. One is the set of all 6994 prompts produced by the human side of the conversation. The other contains only the initial utterance that starts each of the 3000 conversations since this is a more relevant structure to the attacks we observed in Zou et al. (2023).

The first set of plots shows the human prompts' perplexity, token length frequency distribution, and scatter plot.

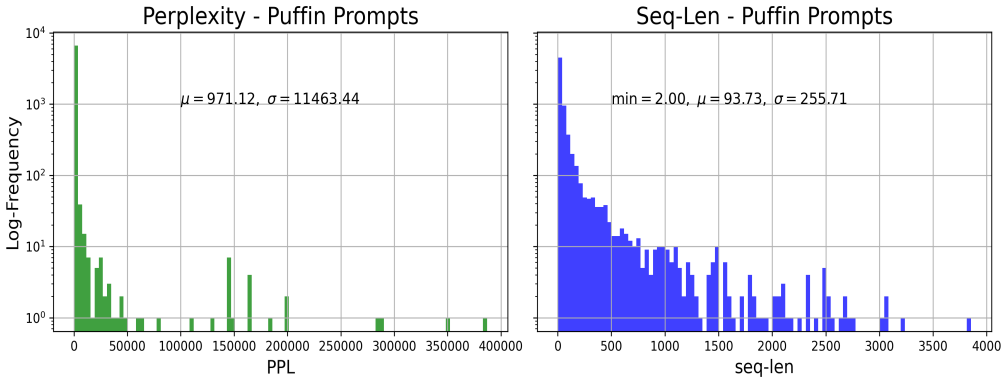

Figure 20: Log-frequency distributions for perplexity and sequence-length

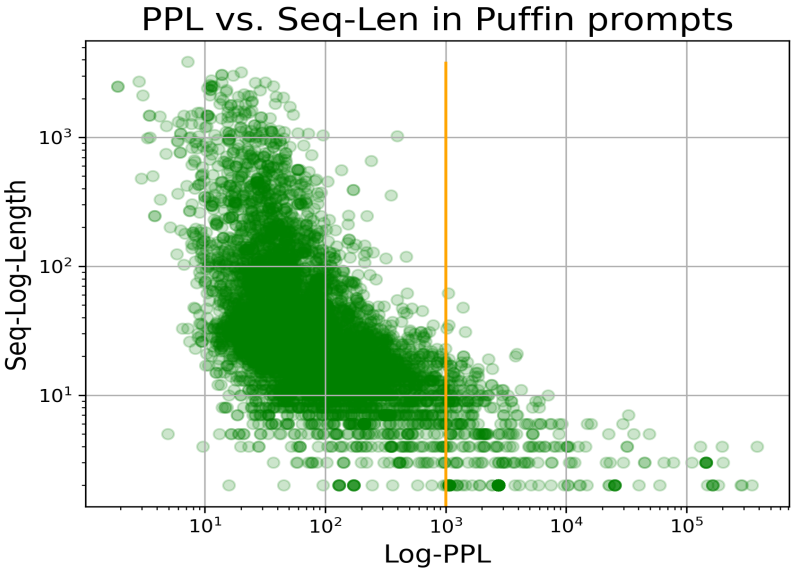

Figure 21: Scatter-plot showing perplexity vs. sequence-length

Some monolectic utterances in Puffin result in "NaN" perplexities. We collected them, and they contain the following items:

'1', '10', '2', '3', '389', '4', '433', '6', '7', 'A', 'B', 'Continue', 'Finish', 'Hi', 'NO', 'No', 'Test', 'Yes', 'a', 'b', 'c', 'conservative', 'continue', 'd', 'hi', 'more', 'next', 'no', 'ok', 'sometimes', 'thanks', 'web', 'yes'

It is easy to conclude that they make little sense standalone, but they are meaningful when considered in the context of a conversation's prior interactions.

The next set of plots shows the perplexity, token length frequency distribution, and scatter plot of the initial human prompt that starts each conversation, Puffin[0].

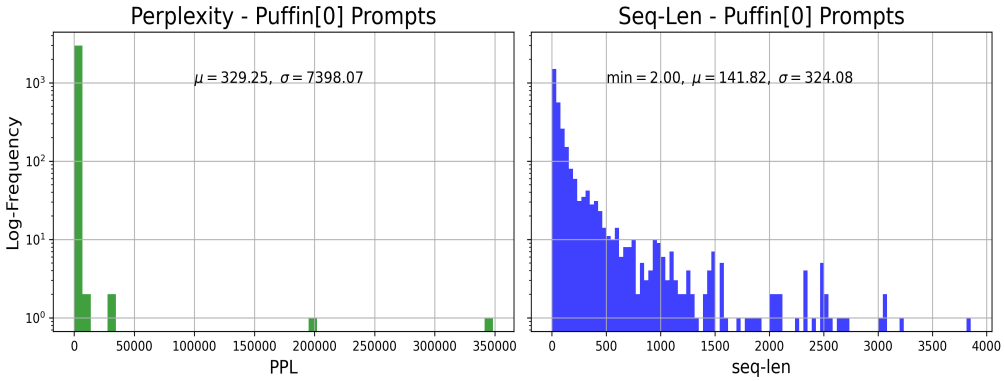

Figure 22: Log-frequency distributions for perplexity and sequence-length

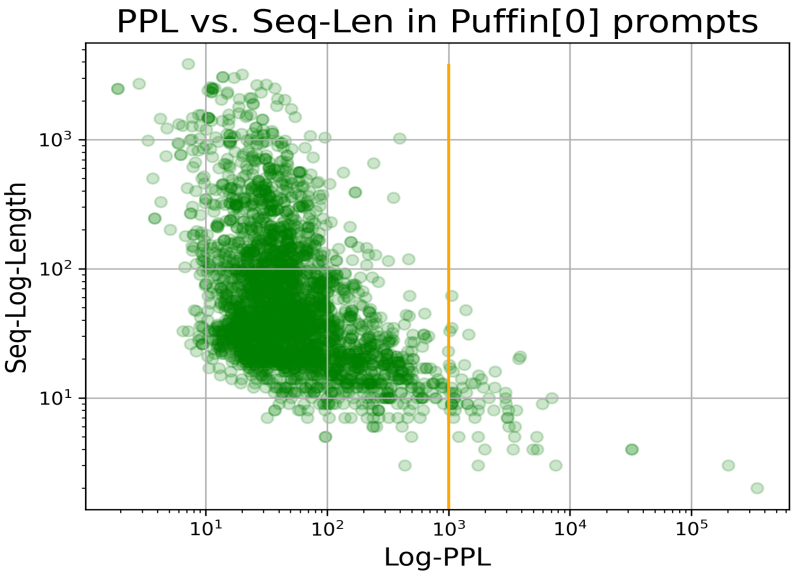

Figure 23: Scatter-plot showing perplexity vs. sequence-length

B.6 TAPIR

This is a large dataset containing examples intended for instruction-following training. We use the Huggingface dataset MattiaL/tapir-cleaned-116k (Mattia Limone, 2023) containing 116862 examples. We construct prompts by concatenating the instruction field and the input field from each example.

The plots below show these prompts' perplexity, token length frequency distribution, and scatter plot.

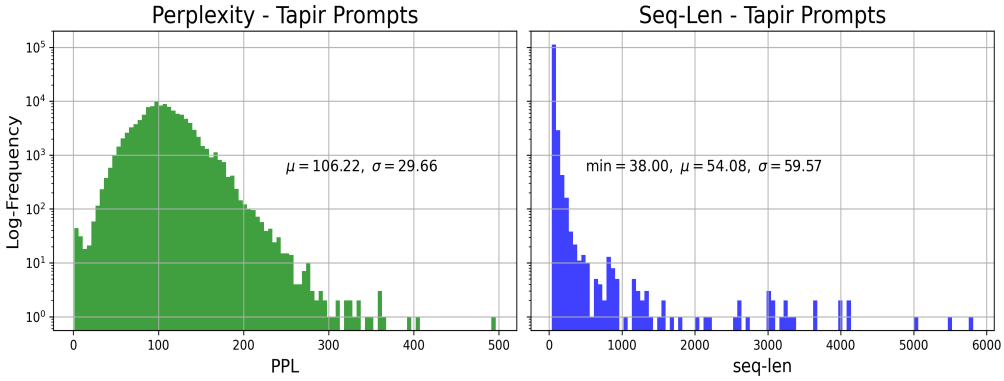

Figure 24: Log-frequency distributions for perplexity and sequence-length

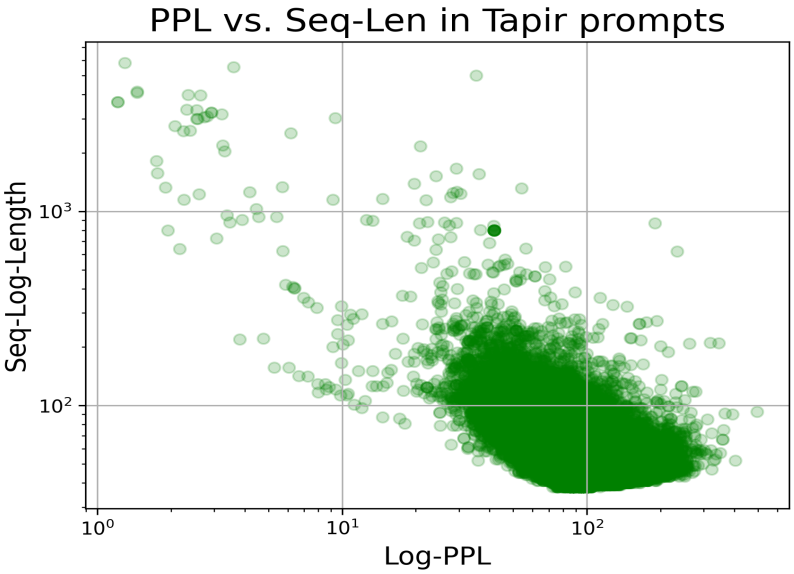

Figure 25: Scatter-plot showing perplexity vs. sequence-length